Topic 2 — Mechanics, Gravitation and Physical Properties of Matter

Direct Syllabus Mapping: General Science — Mechanics (Mass, Weight, Kinematics, Newton's Laws of Motion, Momentum, Work, Power, Energy, Friction, Circular Motion), Gravitation (Kepler's Laws, Universal Gravitation, Acceleration due to Gravity g, Satellites, Escape Velocity), Oscillations (Simple Harmonic Motion, Pendulums), and Physical Properties of Matter (Elasticity, Plasticity, Hooke's Law, Moduli of Elasticity, Surface Tension, Cohesion & Adhesion, Capillarity, Viscosity, Hydrostatic Pressure, Torricelli's Law, Archimedes' Principle, Laws of Floatation, Relative Density).
Key Must-Score Focus: Marginal friction numericals (71st BPSC 2025), Force on truck & work done by oxen (71st BPSC 2025), Bulk Modulus of rigid body = ∞ (UKPCS 2025), Thermal expansion dimensions K^{-1} (UKPCS 2025), Stretching of spring coil determined by Shear Modulus (UP RO/ARO 2023), Gas liquefaction conditions (70th BPSC 2024), Moon lack of atmosphere ($V_e < v_{\text{rms}}$), Elevator apparent weight, Iceberg floating ratios, Water density anomalies at 4°C ($277\ \text{K}$), Surface tension phenomena (soap bubble excess pressure, rain droplets, oil on water, needle floating), Capillarity examples, Viscosity trends with temperature, and Pendulum seasonal/altitude variations.

Current Affairs

- **Confirmed Moons Count in Solar System (2026 IAU Data)**:
- **Saturn**: Maximum number of confirmed satellites in the solar system = **285 moons** (leading the solar system).
- **Jupiter**: **101 confirmed moons** (Ganymede, Callisto, Io, Europa).
- **Uranus**: 29 moons; **Neptune**: 16 moons; **Mars**: 2 moons (Phobos & Deimos); **Earth**: 1 moon; **Mercury & Venus**: 0 moons; Dwarf planet **Pluto**: 5 moons.
- **71st BPSC (Pre 2025) Mechanics & Fluid Facts**:
- *Marginal Kinetic Friction*: Pushing a wooden box with $200\ \text{N}$ force at constant velocity ($a = 0$) $\implies$ Marginal frictional force = $200\ \text{N}$ (net force is zero).
- *Truck Force*: A 7-ton truck ($7,000\ \text{kg}$) covers $400\ \text{m}$ from rest in $20\ \text{s} \implies a = 2\ \text{m/s}^2 \implies F = ma = 14,000\ \text{N}$.
- *Oxen Work*: A pair of oxen exerts $140\ \text{N}$ over a $15\ \text{m}$ field $\implies W = 140 \times 15 = 2100\ \text{Joules}$.
- *Energy Conservation in Free Fall*: Potential energy decreases while kinetic energy increases; total mechanical energy is strictly conserved.
- **UKPCS (Pre 2025) Elasticity & Thermal Facts**:
- *Bulk Modulus for Perfectly Rigid Body*: **Infinite (∞)**. Since volumetric strain is zero ($\Delta V = 0$), $K = -\Delta P/(\Delta V/V) = \infty$.
- *Coefficients of Thermal Expansion*: Linear (α), Superficial (β), and Volumetric (γ) expansion coefficients all have dimensions K^{-1} ($[M^0L^0T^0K^{-1}]$).
- **UP RO/ARO (Pre 2023) Moduli of Elasticity Facts**:

- *Young's Modulus*: Steel has a **higher Young's Modulus than rubber** (Steel is more elastic than rubber).
- *Stretching of a Helical Coil*: Governed by **Shear Modulus (Modulus of Rigidity)** because pulling a coil alters the shape of the wire without changing its length or volume.
- **70th BPSC (Pre 2024) Gas Liquefaction Condition**:
- Gases are liquefied under **low temperature and high pressure** (cooling reduces kinetic energy of molecules while compression forces them into intermolecular attraction range).

Teaching Cards — Mechanics and Properties of Matter

2.1 Motion — Distance, Speed and Acceleration

Kinematics describes motion using distance, displacement, speed, velocity and acceleration without naming the force that causes it.

- **Distance** is the actual path length; it is a scalar and is never negative.
- **Displacement** is the shortest straight-line change of position; it is a vector and can be zero after a round trip.
- **Speed** is distance per unit time; **velocity** is displacement per unit time and has direction.
- For equal distance halves at speeds v_1 and v_2, average speed is the harmonic mean $\frac{2v_1v_2}{v_1+v_2}$.
- Under constant acceleration, $v = u + at$, $s = ut + \frac{1}{2}at^2$, and $v^2 = u^2 + 2as$.
- In free fall from rest (neglecting air), $v = \sqrt{2gh}$ and all masses fall with the same g in vacuum.

2.2 Newton's Laws and Momentum

Newton's laws link force to the change of motion of a body.

- The first law is inertia: a body keeps its state of rest or uniform motion unless a net force acts.
- A passenger leans back when a bus starts and leans forward on sudden braking because of inertia.
- The second law is $F = \frac{dp}{dt} = ma$; impulse $F\Delta t$ equals change of momentum.
- A goalkeeper pulls the hands back while catching to increase Δt and reduce impact force.
- The third law says action and reaction are equal, opposite, and act on **two different bodies**.
- On frictionless ice, blowing air forward produces a backward reaction that moves the body.

2.3 Friction and Circular Motion

Friction opposes relative motion at contact; circular motion needs a centre-seeking force.

- Hierarchy: **static friction** $>$ **kinetic friction** $>$ **rolling friction**.
- A loaded cart is harder to start than to keep moving because limiting static friction is larger.
- Ball bearings convert sliding friction into rolling friction by reducing contact area.
- At constant velocity, net force is zero, so applied force equals kinetic friction.

- **Centripetal force** mv^2/r is real and inward; **centrifugal force** is the outward inertial force in a rotating frame.
- On an unbanked curve, excess speed makes the vehicle skid outward and can overturn about the outer wheels.

2.4 Work, Power and Energy

Work transfers energy by a force acting through a displacement.

- $W = Fs\cos\theta$; work is zero when force is perpendicular to displacement.
- Power is $P = W/t = Fv$; $1\ \text{hp} = 746\ \text{W}$.
- Kinetic energy is $\frac{1}{2}mv^2$; gravitational potential energy near Earth is mgh.
- Mechanical energy is conserved when non-conservative work (like friction heat loss) is absent.
- Simple machines let you use a smaller effort for the same work; they do not create energy.

2.5 Gravitation, Satellites and Escape

Gravitation is the mutual attraction between masses, governed by Newton's inverse-square law.

- Kepler's first law: planets move in ellipses with the Sun at one focus.
- Kepler's second law: areal velocity is constant — fastest at perihelion, slowest at aphelion.
- Kepler's third law: $T^2 \propto a^3$.
- $g = GM/R^2$ is independent of the falling object's mass; it is larger at poles than at the equator.
- Orbital speed near Earth is about $7.92\ \text{km/s}$; escape speed is about $11.2\ \text{km/s} = \sqrt{2}\,V_o$.
- The Moon lacks a lasting atmosphere because lunar escape speed is lower than the typical thermal speed of light gases.
- In an elevator, apparent weight is $m(g + a)$ when accelerating up and $m(g - a)$ when accelerating down; free fall gives weightlessness.

2.6 Pendulum and Simple Harmonic Motion

A simple pendulum approximates SHM for small angles.

- Period $T = 2\pi\sqrt{l/g}$ does **not** depend on bob mass.
- Longer length or smaller g increases period.
- In summer a metal rod expands, l rises, clocks with pendulums **lose** time; in winter they **gain** time.
- A seconds pendulum has $T = 2\ \text{s}$ and length about $1\ \text{m}$ on Earth.

2.7 Elasticity and Moduli

Elasticity is the ability to regain original shape after the deforming force is removed.

- Hooke's law: within the proportional limit, stress is proportional to strain.
- Stress $= F/A$ (pascal); strain $= \Delta L/L$ is dimensionless.
- **Steel is more elastic than rubber** because steel has a much higher Young's modulus.

- Bulk modulus of a perfectly rigid body is **infinite** because volumetric strain is zero.
- Stretching a helical spring coil is governed mainly by the **shear modulus**.

2.8 Surface Tension and Capillarity

Surface tension makes a liquid surface behave like a stretched membrane.

- Drops and bubbles become spherical to minimise surface area.
- Surface tension falls with rising temperature and is zero at the critical temperature.
- Excess pressure for a liquid drop is $2T/R$; for a soap bubble (two surfaces) it is $4T/R$.
- When two soap bubbles connect, air flows from the smaller (higher pressure) to the larger.
- **Cohesion** is like-with-like attraction; **adhesion** is unlike-with-unlike attraction.
- Water rises in a clean glass capillary; mercury shows capillary depression.
- Oil on water spreads because water's surface tension is higher; a straw drink works by suction, **not** capillarity.

2.9 Viscosity and Fluid Statics

Viscosity is internal friction between fluid layers; fluid statics covers pressure and buoyancy.

- In streamlined pipe flow, speed is zero at the wall and maximum on the axis.
- Liquid viscosity **decreases** with temperature; gas viscosity **increases** with temperature.
- Relative viscosity order: honey $>$ glycerine $>$ blood $>$ water $>$ air.
- Pascal's law: pressure in a confined liquid transmits undiminished — basis of hydraulic brakes and presses.
- Torricelli's law: efflux speed at depth h is $\sqrt{2gh}$.
- Archimedes' principle: upthrust equals weight of fluid displaced.
- A floating ice cube melting in water leaves the water level **unchanged**.
- A ship rising from river to sea floats slightly higher because seawater is denser.

Master Tables & Core Formulations

Table 1: Master Formula Matrix for Kinematics & Mechanics

Concept / Law	Formula / Relationship	Key Points
Kinematic Equations (Uniform Acceleration)	$v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$	For free fall under gravity from rest ($u = 0$): $v = gt$, $h = \frac{1}{2}gt^2$, $v = \sqrt{2gh}$.
Displacement in n^{th} second	$s_n = u + \frac{a}{2}(2n - 1)$	Ratio of distances in successive seconds from rest: $1 : 3 : 5 : 7 : \ldots$ (Galileo's odd-number rule).
Harmonic Average Speed	$v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2}$	Applies when two equal distances are traversed at speeds v_1 and v_2.

Concept / Law	Formula / Relationship	Key Points
Newton's 2nd Law & Momentum	$F = \frac{dp}{dt} = ma = m\frac{v-u}{t}$	Impulse $J = F\Delta t = \Delta p = m(v-u)$ ($N \cdot s$).
Work Done	$W = Fs\cos\theta$	$W > 0$ if $\theta < 90°$; $W = 0$ if $\theta = 90°$ (centripetal force); $W < 0$ if $\theta = 180°$ (friction).
Power	$P = \frac{W}{t} = Fv$	$1\ \text{W} = 1\ \text{J/s}$; $1\ \text{hp} = 746\ \text{W}$.
Kinetic Energy & Momentum	$KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$	If momentum p is doubled, KE becomes 4 times.
Gravitational Potential Energy	$PE = mgh$	Stored energy due to position in Earth's gravitational field.
Universal Gravitation	$F = G\frac{m_1 m_2}{r^2}$	$G = 6.67430 \times 10^{-11}\ \text{N} \cdot m^2/kg^2$ (measured by Cavendish).
Acceleration due to Gravity	$g = \frac{GM_e}{R_e^2}$	Independent of mass of falling object.
Escape Velocity	$V_e = \sqrt{\frac{2GM_e}{R_e}} = \sqrt{2gR_e}$	$V_e = \sqrt{2}V_o \approx 11.2\ \text{km/s}$ on Earth; $\approx 2.38\ \text{km/s}$ on Moon.
Simple Pendulum Period	$T = 2\pi\sqrt{\frac{l}{g}}$	$T \propto \sqrt{l}$; $T \propto 1/\sqrt{g}$. Independent of mass m.
de Broglie Wavelength	$\lambda = \frac{h}{p} = \frac{h}{mv}$	$h = 6.626 \times 10^{-34}\ \text{J} \cdot s$. Connects particle mass-velocity to wave nature.

Table 2: Planetary Satellites Master Directory (2026 Astronomy Consensus)

Celestial Body	Number of Confirmed Moons	Key Noteworthy Moons
Saturn	**285** (Maximum in Solar System)	Titan (thick nitrogen atmosphere), Enceladus (cryovolcanoes/water ocean), Mimas, Rhea
Jupiter	**101**	Ganymede (largest moon in solar system), Callisto, Io (volcanic), Europa (subsurface ocean)
Uranus	**29**	Titania, Oberon, Miranda, Ariel, Umbriel
Neptune	**16**	Triton (retrograde orbit, nitrogen geysers), Proteus, Nereid
Mars	**2**	Phobos, Deimos (captured asteroids)
Earth	**1**	Moon (Luna)
Venus & Mercury	**0**	No natural satellites
Pluto (Dwarf)	**5**	Charon (tidally locked binary system), Styx, Nix, Kerberos, Hydra

Table 3: Comprehensive Energy Transformations Matrix

Apparatus / System	Input Energy Form	Output Energy Form	Everyday / Industrial Example
Dynamo / Alternator	Mechanical	Electrical	Bicycle generator, hydroelectric turbine
Electric Motor	Electrical	Mechanical	Ceiling fan, water pump, electric vehicle
Microphone	Acoustic (Sound)	Electrical	Audio recording, telephone transmitters
Loudspeaker	Electrical	Acoustic (Sound)	Public address system, headphones
Solar Photovoltaic Cell	Radiant (Light)	Electrical	Solar rooftop panels, solar calculators
Tubelight / LED	Electrical	Radiant (Light)	Household room illumination
Incandescent Bulb	Electrical	Thermal (Heat) + Light	Tungsten filament lamp (90% heat, 10% light)
Chemical Primary Cell	Chemical	Electrical	Dry cell battery in remotes/torches
Candle	Chemical	Thermal (Heat) + Light	Paraffin wax combustion
Sitar / Guitar	Mechanical	Acoustic (Sound)	Plucking vibrating taut strings
Photosynthesis	Solar (Light)	Chemical	Chlorophyll in green plants synthesizing glucose
Automobile Brakes	Kinetic	Thermal (Heat)	Friction pads slowing wheel rotation
Nuclear Reactor	Nuclear Mass Energy	Thermal (Heat)	Fission of Uranium-235 generating steam ($E = mc^2$)

Table 4: Mechanical Properties of Solids & Fluids Master Formulations

Quantity / Law	Mathematical Expression	SI Unit / Dimensions	Must-Score Core Fact
Hooke's Law	$\text{Stress} = E \times \text{Strain}$	N/m^2 or Pa	Valid within proportional elastic limit.
Young's Modulus (Y)	$Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$	N/m^2 ($[M^1L^{-1}T^{-2}]$)	Longitudinal elasticity; $Y_{\text{steel}} > Y_{\text{rubber}}$.
Shear Modulus (η)	$\eta = \frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{F/A}{\theta}$	N/m^2 ($[M^1L^{-1}T^{-2}]$)	Determines stretching and twisting of helical springs.
Bulk Modulus (K)	$K = -\frac{\Delta P}{\Delta V/V}$	N/m^2 ($[M^1L^{-1}T^{-2}]$)	For perfectly rigid solid, $\Delta V = 0 \implies K = \infty$. Compressibility $\beta = 1/K$.

Quantity / Law	Mathematical Expression	SI Unit / Dimensions	Must-Score Core Fact
Thermal Expansion	$\Delta L = \alpha L \Delta T$; $\Delta V = \gamma V \Delta T$	α, β, γ have unit K^{-1}	$[M^0 L^0 T^0 K^{-1}]$; $\alpha : \beta : \gamma = 1 : 2 : 3$.
Surface Tension (T)	$T = \frac{F}{L} = \frac{\Delta W}{\Delta A}$	N/m or J/m^2 ($[M^1 L^0 T^{-2}]$)	Decreases with temperature; zero at critical point.
Excess Pressure (Bubble)	$\Delta P = \frac{4T}{R}$	Pa or N/m^2	Smaller bubble has higher internal pressure.
Capillary Rise (h)	$h = \frac{2T\cos\theta}{\rho g r}$	Metres (m)	Jurin's Law ($h \propto 1/r$); acute $\theta \implies$ rise; obtuse $\theta \implies$ fall.
Viscous Force (F)	$F = -\eta A \frac{dv}{dx}$ (Newton's Law)	η in $Pa \cdot s$ or $Poise$	$1\ \text{Pa} \cdot s = 10\ \text{Poise}$. Poiseuille's formula for pipe flow.
Torricelli's Efflux	$v = \sqrt{2gh}$	m/s	Velocity of liquid flowing out of an orifice at depth h.
Hydrostatic Pressure	$P = P_0 + \rho g h$	N/m^2 or Pa	Increases linearly with liquid depth; independent of container shape.
Archimedes' Buoyancy	$F_B = \rho_{\text{fluid}} V_{\text{submerged}} g$	Newton (N)	Acting vertically upward through centre of buoyancy.
Floatation Fraction	$\frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{solid}}}{\rho_{\text{liquid}}}$	Dimensionless ratio	Determines how deep an iceberg or ship sinks.

Table 5: Buoyancy, Density & Floatation Phenomena Master Matrix

Phenomenon / System	Underlying Physical Mechanism	Must-Score Consequence & Rule
Iron nail sinks, iron ship floats	Ship hull encloses vast air volume, lowering overall average density below water; displaced water weight equals ship weight.	Archimedes' Principle; solid iron density ($7.8\ \text{g/cm}^3$) > water ($1.0\ \text{g/cm}^3$).
Iron ball on Mercury vs Water	Density hierarchy: $\rho_{\text{water}}(1.0) < \rho_{\text{iron}}(7.8) < \rho_{\text{mercury}}(13.6)$.	Iron sinks in water but floats on mercury.
Melting floating ice in a glass	Ice displaces water equal to its mass ($m = \rho_w V_w = \rho_i V_i$). When melted, it turns into exact volume V_w.	Water level remains completely unchanged.
Ship moving from river to ocean	Sea water has dissolved salts $\implies$ higher density ($1.025\ \text{g/cm}^3$) $\implies$ higher buoyant upthrust.	Ship rises slightly when entering the sea.
Swimming in sea vs river	Higher density of sea water provides greater buoyant force per unit submerged volume.	Much easier to swim and stay afloat in sea water.

Phenomenon / System	Underlying Physical Mechanism	Must-Score Consequence & Rule
Iceberg in ocean water	Density of pure ice $\approx 0.917\ \text{g/cm}^3$; ocean water $\approx 1.025\ \text{g/cm}^3$.	$\approx 1/9\text{th}$ of iceberg is above sea surface (89% submerged).
Needle resting on water surface	Upward component of surface tension along boundary of surface film supports the small weight.	Surface tension (NOT buoyancy; if pushed below surface, needle sinks).
Rising air bubble in water	Hydrostatic pressure decreases as bubble ascends ($P = P_0 + \rho g h$). By Boyle's Law ($PV = \text{const}$), volume expands.	Bubble increases in size as it reaches the surface.
Leaking fountain pen in airplane	Atmospheric cabin pressure drops at high altitude while ink reservoir pressure remains high.	Ink is forced out of the pen nib.

Bilingual Terminology -- Mechanics and Properties of Matter

English Term	Hindi Term	Key Anchor
Force (बल)	बल	Push or pull causing acceleration; SI unit = Newton (N)
Newton's Laws (न्यूटन के नियम)	गति के नियम	3 laws governing motion and force relationships
Inertia (जड़त्व)	जड़त्व	Tendency of object to resist change in its state of motion
Momentum (संवेग)	संवेग	Mass x Velocity; p = mv; conserved in absence of external force
Friction (घर्षण)	घर्षण	Resistive force between surfaces in contact
Gravity / Gravitation (गुरुत्वाकर्षण)	गुरुत्वाकर्षण	Attractive force between masses; Newton's Law of Gravitation
Weight (भार)	भार	Gravitational force on object; W = mg; varies with location
Mass (द्रव्यमान)	द्रव्यमान	Amount of matter; constant everywhere; SI unit = kg
Pressure (दाब)	दाब	Force per unit area; P = F/A; SI unit = Pascal (Pa)
Buoyancy (उत्प्लावकता)	उत्प्लावकता	Upward force on object immersed in fluid; Archimedes' Principle
Archimedes' Principle (आर्किमिडीज का सिद्धांत)	आर्किमिडीज का नियम	Upthrust = weight of fluid displaced
Pascal's Law (पास्कल का नियम)	पास्कल का नियम	Pressure applied to enclosed fluid = transmitted equally in all directions
Bernoulli's Principle (बर्नूली का सिद्धांत)	बर्नूली का सिद्धांत	Faster fluid flow = lower pressure; explains flight, carburetor, venturimeter

English Term	Hindi Term	Key Anchor
Surface Tension (पृष्ठ तनाव)	पृष्ठ तनाव	Tendency of liquid surface to contract; insects walk on water
Viscosity (श्यानता)	श्यानता	Resistance of liquid to flow; honey > water; decreases with temp
Elasticity (प्रत्यास्थता)	प्रत्यास्थता	Ability to return to original shape after deforming force removed
Hooke's Law (हुक का नियम)	हुक का नियम	Stress proportional to strain within elastic limit
Torque (बल-आघूर्ण)	बल-आघूर्ण	Rotational equivalent of force; T = F x r (moment arm)

One-Liner Revision -- Mechanics and Properties of Matter

#	Fast Fact
1	Newton's 1st Law = Law of **Inertia** (object at rest stays at rest)
2	Newton's 2nd Law = F = ma (Force = mass x acceleration)
3	Newton's 3rd Law = For every action, there is equal and opposite **reaction**
4	1 Newton = force that accelerates 1 kg mass by 1 m/s2
5	Momentum (p) = m x v; Law of Conservation of Momentum
6	Impulse = Force x Time = change in momentum
7	g (acceleration due to gravity) = 9.8 m/s2 on Earth's surface
8	g is maximum at poles; minimum at equator (due to Earth's shape)
9	g decreases with altitude and depth (both)
10	Weight = 0 in free fall / space (apparent weightlessness)
11	Escape velocity from Earth = **11.2 km/s**
12	Orbital velocity (LEO) = ~7.9 km/s
13	Archimedes' Principle: Upthrust = weight of fluid displaced
14	A body floats if its average density <= density of fluid
15	Density of water = 1000 kg/m3 (1 g/cm3); ice floats (density ~917 kg/m3)
16	Pascal's Law: basis of hydraulic press, brakes, jacks
17	Bernoulli's Principle: explains airplane wing lift, atomizer, carburetor
18	Surface tension: water meniscus, capillarity, insects on water, dew drops
19	Viscosity decreases with temperature (liquids); increases for gases

#	Fast Fact
20	Rolling friction < Sliding friction < Static friction
21	Friction is useful: walking, brakes, writing; harmful: wear of machine parts
22	Centre of Gravity = point where entire weight appears to act
23	Centripetal force = mv2/r (directed towards centre of circular motion)
24	Centrifuge separates components based on density difference (cream from milk)
25	Hooke's Law: stress = E x strain (E = Young's modulus); within elastic limit
26	Elasticity: rubber > steel? No! Steel has **higher** Young's modulus (more elastic by physics definition)
27	Pressure at depth h in liquid: P = rho x g x h
28	Ball pen works on: capillary action + gravity
29	Geostationary orbit = 35,786 km above equator; orbital period = 24 hours
30	Kepler's 3rd Law: T2 is proportional to a3 (period2 proportional to semi-major axis3)